Meeting 1126

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November 26, 2025

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Research motivation

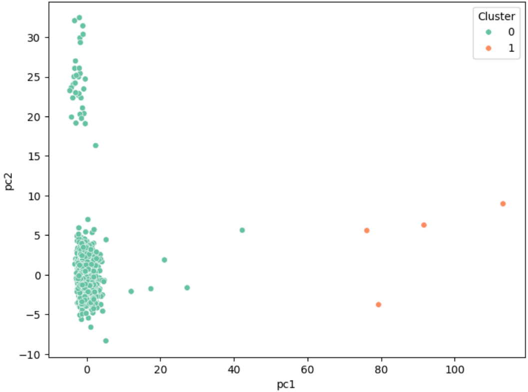
Semiconductor Manufacturing Process

- Unbox $y = \hat{f}(x)$
- Timing to use SMOTE (deal with imbalanced data)
- Fitting model

Research method

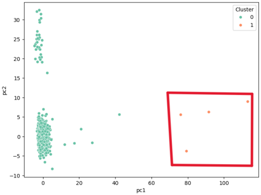
- Dimensionality reduction
  - PCA
- Clustering
  - KMeans
  - AgglomerativeClustering(Hierarchical clustering)
- Model training(StandardScaler, SMOTE on training dataset)
  1. LR, SVM, DT, RF, XGB
  2. ACC, FAR

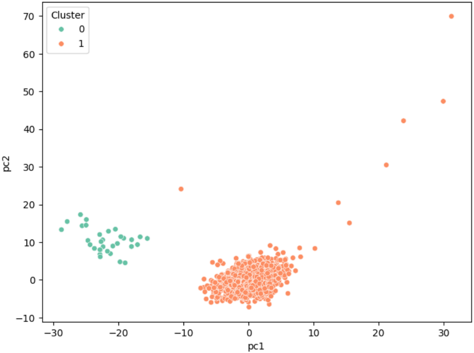
Results I PCA + KMeans, 2 clusters



cluster	0	1
$y = 0$	1460	3
$y = 1$	103	1

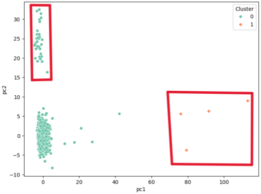
Results II PCA + KMeans, 2 clusters

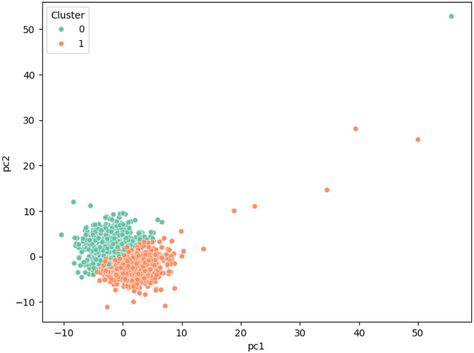




cluster	0	1
$y=0$	1435	25
$y=1$	96	7

Results III PCA + KMeans, 2 clusters





cluster	0	1
$y=0$	669	765
$y=1$	60	36

Results III Training in cluster 0

LR

$$\begin{bmatrix} 122 & 12 \\ 10 & 2 \end{bmatrix}$$

SVM

$$\begin{bmatrix} 133 & 1 \\ 12 & 0 \end{bmatrix}$$

DT

$$\begin{bmatrix} 113 & 21 \\ 8 & 4 \end{bmatrix}$$

RF

$$\begin{bmatrix} 134 & 0 \\ 12 & 0 \end{bmatrix}$$

XGB

$$\begin{bmatrix} 130 & 4 \\ 12 & 0 \end{bmatrix}$$

Summary

model	ACC	FAR
LR	0.8493	0.0896
SVM	0.9110	0.0075
DT	0.8014	0.1567
RF	0.9178	0.0000
XGB	0.8904	0.0299

Results III Training in cluster 1

LR

$$\begin{bmatrix} 149 & 6 \\ 5 & 2 \end{bmatrix}$$

SVM

$$\begin{bmatrix} 154 & 0 \\ 7 & 0 \end{bmatrix}$$

DT

$$\begin{bmatrix} 141 & 13 \\ 5 & 2 \end{bmatrix}$$

RF

$$\begin{bmatrix} 154 & 0 \\ 7 & 0 \end{bmatrix}$$

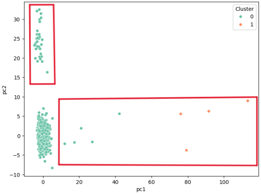
XGB

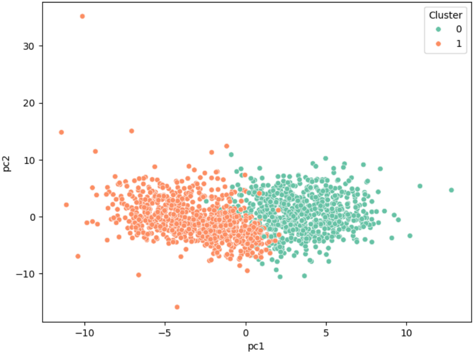
$$\begin{bmatrix} 153 & 1 \\ 7 & 0 \end{bmatrix}$$

Summary

model	ACC	FAR
LR	0.9379	0.0325
SVM	0.9565	0.0000
DT	0.8882	0.0844
RF	0.9565	0.0000
XGB	0.9503	0.0065

Results IV PCA + KMeans, 2 clusters





cluster	0	1
$y = 0$	660	770
$y = 1$	58	37

Results IV Training in cluster 0

LR

$$\begin{bmatrix} 114 & 18 \\ 6 & 6 \end{bmatrix}$$

SVM

$$\begin{bmatrix} 131 & 1 \\ 12 & 0 \end{bmatrix}$$

DT

$$\begin{bmatrix} 117 & 15 \\ 10 & 2 \end{bmatrix}$$

RF

$$\begin{bmatrix} 132 & 0 \\ 12 & 0 \end{bmatrix}$$

XGB

$$\begin{bmatrix} 129 & 3 \\ 12 & 0 \end{bmatrix}$$

Summary

model	ACC	FAR
LR	0.8333	0.1364
SVM	0.9097	0.0076
DT	0.8264	0.1136
RF	0.9167	0.0000
XGB	0.8958	0.0227

Results IV Training in cluster 1

LR

$$\begin{bmatrix} 141 & 14 \\ 6 & 1 \end{bmatrix}$$

SVM

$$\begin{bmatrix} 155 & 0 \\ 7 & 0 \end{bmatrix}$$

DT

$$\begin{bmatrix} 136 & 19 \\ 6 & 1 \end{bmatrix}$$

RF

$$\begin{bmatrix} 155 & 0 \\ 7 & 0 \end{bmatrix}$$

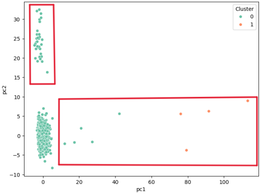
XGB

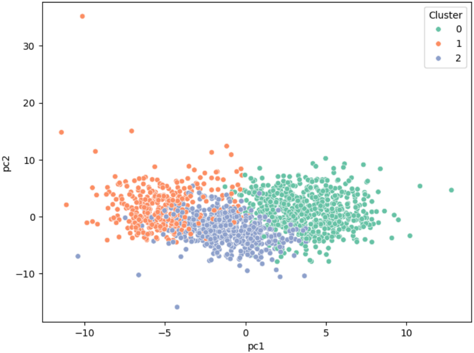
$$\begin{bmatrix} 155 & 0 \\ 7 & 0 \end{bmatrix}$$

Summary

model	ACC	FAR
LR	0.8765	0.0903
SVM	0.9568	0.0000
DT	0.8457	0.1226
RF	0.9568	0.0000
XGB	0.9568	0.000

Results V PCA + KMeans, 3 clusters





cluster	0	1	2
$y = 0$	592	327	511
$y = 1$	49	19	27

Results V Training in cluster 0

LR SVM DT

$$\begin{bmatrix} 100 & 19 \\ 6 & 4 \end{bmatrix} \quad \begin{bmatrix} 119 & 0 \\ 10 & 0 \end{bmatrix} \quad \begin{bmatrix} 99 & 20 \\ 8 & 2 \end{bmatrix}$$

RF XGB

$$\begin{bmatrix} 118 & 1 \\ 10 & 0 \end{bmatrix} \quad \begin{bmatrix} 115 & 4 \\ 10 & 0 \end{bmatrix}$$

Summary

model	ACC	FAR
LR	0.8765	0.0903
SVM	0.9568	0.0000
DT	0.8457	0.1226
RF	0.9568	0.0000
XGB	0.9568	0.000

The End

Questions & Comments